Differential Manifolds

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Contents

Chapter I

§1

Given a funtion $f(x,y)$ defined on a deleted neighborhood of (x_0, y_0)

Proposition 1.1. *If* $\lim_{(x,y)\to(x_0,y_0)} f(x,y) = A$

1. *if* $\lim_{x\to x_0} f(x,y)$ *exists for all* $y \neq y_0$, *then* $\lim_{y\to y_0} \lim_{x\to x_0} f(x,y) = A$.
2. *if* $\lim_{y\to y_0} f(x,y)$ *exists for all* $x \neq x_0$, *then* $\lim_{x\to x_0} \lim_{y\to y_0} f(x,y) = A$.

Example 1.2.

$$f(x,y) = \begin{cases} x \sin \frac{1}{y} & , y \neq 0 \\ 0 & , y = 0 \end{cases}$$

$\lim_{(x,y)\to(0,0)} f(x,y) = \lim_{y\to 0} \lim_{x\to 0} f(x,y) = 0$, *but* $\lim_{x\to 0} \lim_{y\to 0} f(x,y)$ *does not exist.*

Example 1.3.

$$f(x,y) = \frac{x^2 - y^2}{x^2 + y^2}$$

Proposition 1.4. *If*

(i) $\lim_{y\to y_0} f(x,y) = g(x)$ *exists for all* $x \neq x_0$

(ii) $\lim_{x\to x_0} f(x,y) = h(y)$ *uniformly for all* $y \neq y_0$

then $\lim g(x) = \lim h(y)$ *exists and* $\lim g(x) = \lim h(y)$.